



Research paper



Region-specific genomic signatures of multidrug-resistant *Helicobacter pylori* isolated from East and South India

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ABSTRACT

Helicobacter pylori is a ubiquitous bacterium and contributes significantly to the burden of chronic gastritis, peptic ulcers, and gastric cancer across the world. Adaptive phenotypes and virulence factors in *H. pylori* are heterogeneous and dynamic. However, limited information is available about the molecular nature of antimicrobial resistance phenotypes and virulence factors of *H. pylori* strains circulating in India. In the present study, we analyzed the whole genome sequences of 143 *H. pylori* strains, of which 32 are isolated from two different regions (eastern and southern) of India. Genomic repertoires of individual strains show distinct region-specific signatures. We observed lower resistance phenotypes and genotypes in the East Indian (Kolkata) *H. pylori* isolates against amoxicillin and furazolidone antibiotics, whereas higher resistance phenotypes to metronidazole and clarithromycin. Also, at molecular level, a greater number of AMR genes were observed in the east Indian *H. pylori* isolates as compared to the southern Indian isolates. From our findings, we suggest that metronidazole and clarithromycin antibiotics should be used judiciously in the eastern India. However, no horizontally acquired antimicrobial resistance gene was observed in the current *H. pylori* strains. The comparative genome analysis shows that the number of genes involved in virulence, disease and resistance of *H. pylori* isolated from two different regions of India is significantly different. Single-nucleotide polymorphisms (SNPs) based phylogenetic analysis distinguished *H. pylori* strains into different clades according to their geographical locations. Conditionally beneficial functions including antibiotic resistance phenotypes that are linked with faster evolution rates in the Indian isolates.

Abbreviations: ABC, ATP binding cassette transporter; AMR, Antimicrobial resistance; BHI, Brain Heart Infusion; *CagA*, Cytotoxin associated gene A; CARD, Comprehensive Antibiotic Resistance Database; CDC, Centers for Disease Control and Prevention; EUCAST, European Committee on Antimicrobial Susceptibility Testing; MALT, Mucosa-associated lymphoid tissue; MATE, Multidrug and toxic compound extrusion; MFS, Major facilitator superfamily; MGE, Mobile genetic elements; PAI, Pathogenicity island; PBS, Phosphate Buffered Saline; RAST, Rapid Annotation using Subsystem Technology; SNP, Single-nucleotide polymorphisms; *VacA*, Vacuolating cytotoxin A; RND, Resistance nodulation cell division.

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1. Introduction

Helicobacter pylori is a high-priority bacterial pathogen and is considered to be one of the leading causative agent of gastric cancer, gastric mucosa-associated lymphoid tissue (MALT) lymphoma, duodenal and gastric ulcers (WHO, 2017; NIH-NCI, 2013). It is a spiral-shaped bacterium that secretes urease to survive in the acidic environment of the stomach. This enzyme converts urea to ammonia, which neutralizes the stomach acidity allowing its growth. According to the Centers for Disease Control and Prevention (CDC), approximately two-thirds of the world's population harbours this bacterium and it is more prevalent in developing countries with prevalence ranging between 85% and 95% in developing countries and 30% to 50% in developed countries (NIH-NCI 2013; Khoder et al., 2019). It is always a point of debate whether *H. pylori* is commensal or not as it is present in most of the healthy population and the pathogenicity is only attributed to a small population (Reshetnyak et al., 2021). *H. pylori* is reported as a group 1 carcinogen and is a major cause of 95% of all gastric cancers (IARC, 1994; Zhang et al., 2014). Since the eastern and southern regions of India have the four-time higher frequency rates of gastric cancer than Northern region of India, this could be due to comparable eating patterns, which differ from the northern region's wheat-based diet (Misra et al., 2014). *H. pylori* possesses several virulence genes for its survival which directly affects gastric epithelium. These are mainly VacA (vacuolating cytotoxin A) and CagA (cytotoxin associated gene A) (Šterbenc et al., 2019; Maeda et al., 1998). In Kolkata, eastern India, 80 to 90% of 78 clinical isolates of *H. pylori* have cag pathogenicity island (PAI) and vacAs1 alleles of the vacuolating cytotoxin gene (vacA) (Mukhopadhyay et al., 2000), whereas in Chennai, southern India, cagA gene prevalence was 96% (90 out of 94 strains), whereas the prevalence of vacA m2 subtype and vacA m1 subtype were 60% and 32%, respectively. Authors suggest vacA m2 subtype as a dominant *H. pylori* strain in South India mostly at Chennai (Udhayakumar et al., 2009).

Clarithromycin, metronidazole and levofloxacin are some of the antibiotics that are frequently used to treat *H. pylori* infection, but several reports suggest increasing resistance over time in India. Thyagarajan S.P et al., (2003) reported resistance pattern for metronidazole (77.9%), clarithromycin (44.7%) and amoxicillin (32.8%) for samples collected from all over India. Increased metronidazole resistance was reported at Lucknow (68%), Chennai (88.2%) and Hyderabad (100%). Highest prevalence of resistant strains were reported for metronidazole, medium for clarithromycin and amoxicillin, and the lowest for ciprofloxacin and tetracycline. Also, higher resistance was reported for southern India than northern India (Thyagarajan et al., 2003). At Kolkata (eastern India), 57 out of 67 (85%) of the clinical isolates from gastritis and peptic ulcer patients were resistant to metronidazole, 7.5% to tetracycline whereas all strains were susceptible to clarithromycin, furazolidone and amoxicillin (Datta et al., 2005). Similarly, from Karnataka, South India, among 113 *H. pylori* strains assayed resistance pattern reported was metronidazole (81.4%), levofloxacin (54.9%), clarithromycin (20.4%), tetracycline (5.3%) and amoxicillin (7.1%). Again, a highest resistance prevalence was observed for metronidazole and levofloxacin, and medium for clarithromycin (Shetty et al., 2019). The living environment and genetic nature of the resistance phenotypes determine how quickly a resistance evolves, is maintained, and is transmitted to other bacterial species. Advances in genome sequencing techniques provide an opportunity for in-depth genomics of pathogens to understand molecular evolution and dissect virulence factors, resistance functions and other fitness phenotypes linked with microbial adaptation. In this study, we have performed comparative genomic analysis of *H. pylori* strains isolated from India and other regions of the globe to find out (i) Antimicrobial resistance phenotypes of *H. pylori* isolates (n = 32) from eastern and southern regions of India (ii) Genetic nature of virulence factors and antimicrobial resistance functions (iii) Genomic variations of *H. pylori* strains isolated from different regions of the world including India. We compared the whole genome sequences of

our recently reported *H. pylori* strains (n = 12) (Mehrotra et al., 2021) with new isolates (n = 20) collected from eastern India. Findings from the present study will help to understand the real-time evolution and genomic heterogeneity of *H. pylori* strains currently circulating in India. In addition, it will be useful for selecting appropriate antimicrobial regimens to treat *H. pylori* infection throughout India and developing nucleic acid probes for rapid identification of resistance phenotypes.

2. Results

2.1. Mapping of the antimicrobial resistance profile

All *H. pylori* strains (n = 20) isolated from Kolkata (East India) were subjected to antimicrobial susceptibility testing by agar dilution against 6 antibiotics (metronidazole, levofloxacin, amoxicillin, tetracycline, clarithromycin and furazolidone), which are frequently used in anti-*H. pylori* therapy (Table 1). Among these 20 strains, maximum resistance was observed against two antibiotics (i) metronidazole (15 out of 20 strains) and (ii) levofloxacin (14 out of 20 strains). In this scenario, indubitably these antibiotics cannot be the choice of therapeutic antibiotics in the future. Resistance profiles of other antibiotics are (iii) clarithromycin (4 out of 20 strains), (iv) tetracycline (all susceptible), (v) furazolidone (all susceptible), (vi) amoxicillin (1 out of 20 strains). Compared to our previous findings (Mehrotra et al., 2021), we observed lower resistance phenotypes in the East Indian *H. pylori* isolates against amoxicillin, furazolidone and tetracycline antibiotics (Fig. 1).

2.2. Quality assessment of *H. pylori* genome sequences

To understand the genetic nature of antibiotic resistance phenotypes a total of 20 *H. pylori* strains isolated at Kolkata were sequenced on Illumina MiSeq DNA sequencing platform. Total 8,731,921 paired-end reads were generated from 20 strains. To generate the draft genome, average 436596.05 (sd ± 92426.88 for 20 strains) paired-end reads with 234 nucleotide average read length were used in *de novo* genome assembly pipeline. Each genome has approximately 61.25X (sd ± 12.97) sequencing depth. QUAST analysis reveals that assembled genomes of 20 isolates of this study has covered 63.186% to 77.202% of reference genome (Supplementary Table 1). There are around 21 to 35 contigs that are misassembled due to either relocation (different position in reference genome) or orientation though all these contigs belong to *H. pylori* strains. CheckM analysis reveals the completeness of draft genomes of all *H. pylori* strains is ranging between 97.16% and 99.62% (Supplementary Table 1). The genome of a single *H. pylori* isolate A14C (5) has no heterogeneity and the contamination is 0.06%.

2.3. Genome sequence analysis of multidrug-resistant *Helicobacter pylori* isolates

Whole genome sequences of *H. pylori* isolates were analysed to identify antimicrobial resistance (AMR) genes, its abundance and genetic linkage with mobile genetic elements (MGEs). Relevant whole genome sequencing information of *H. pylori* strains were summarized in Supplementary Table 2. Total 19 insertion sequence (IS) elements IS605, IS606, ISCaje3, ISCo7, ISHco2, ISCamsp1, ISHahy12, ISCb2, ISCo3, ISvEsV1_1, ISCo4, ISHp608, ISCo5, ISWosp5, Tn4430, ISBas1, ISMth1, IS607 and ISCysp15 were identified in the genome of the *H. pylori* strains isolated in the present study. The IS elements are present in multiple copies and their number vary widely among different isolates (Supplementary Table 3). The results show the presence of 22 out of 55 AMR genes that are common among these strains (Supplementary Table 4). Our findings suggest the presence of AMR genes against different drug classes described in detail (Fig. 2, Supplementary Table 5). Some of the AMR genes repeating multiple times in a particular strain for example, *macB* (macrolide antibiotic) was observed six times, *tet* (44) (tetracycline antibiotic) and *lpsA* (peptide antibiotic)

Table 1Details of *H. pylori* strains isolated from Kolkata (Eastern India) and their antimicrobial susceptibility pattern.

SN	Sample ID	Source	Patient details			Strain details		Antimicrobial Susceptibility pattern of the strain					
			Date of Isolation	Age/Sex	Disease Status	cagA	vacA	MET	TET	CLR	FUZ	LEV	AMX
1	A3C(4)	Clinical Isolate	10.01.2019	32/F	NUD	+ve	s1m1	R	S	S	S	S	S
2	A4A(4)	Clinical Isolate	10.01.2019	38/F	NUD	-ve	s2m2	S	S	S	S	R	S
3	A10A(5)	Clinical Isolate	16.01.2019	30/M	DU	+ve	s1m1	R	S	S	S	R	S
4	A14C(5)	Clinical Isolate	24.01.2019	63/M	DU	+ve	s1m2	S	S	R	S	R	S
5	A15C(4)	Clinical Isolate	24.01.2019	47/F	NUD	-ve	s2m2	S	S	S	S	R	S
6	A26A(1)	Clinical Isolate	13.02.2019	55/M	DU	-ve	s1m1	R	S	S	S	S	S
7	A41C(4)	Clinical Isolate	13.03.2019	31/M	NUD	+ve	s1m2	R	S	S	S	S	S
8	A45C(1)	Clinical Isolate	20.03.2019	28/M	DU	+ve	s1m1	R	S	S	S	R	S
9	A51C(1)	Clinical Isolate	27.03.2019	49/M	DU	+ve	s1m1	S	S	S	S	R	S
10	A59C(1)	Clinical Isolate	10.04.2019	40/F	FD	+ve	s1m1	S	S	R	S	R	S
11	A61C(1)	Clinical Isolate	24.04.2019	55/F	CA stomach	+ve	s1m1	R	S	S	S	R	R
12	A76C(2)	Clinical Isolate	29.05.2019	15/F	GERD	-ve	s2m2	S	S	S	S	R	S
13	A81C(1)	Clinical Isolate	12.06.2019	35/M	DU	+ve	s2m2	R	S	S	S	R	S
14	A82C(1)	Clinical Isolate	12.06.2019	36/M	NUD	+ve	s1m1	R	S	S	S	R	S
15	A93C(2)	Clinical Isolate	20.06.2019	60/M	FD	+ve	s1m1	R	S	R	S	R	S
16	A113A(1)	Clinical Isolate	31.07.2019	45/M	DU	+ve	s1m2	R	S	R	S	R	S
17	A138C(1)	Clinical Isolate	16.10.2019	52/M	Benign Gastric Ulcer	+ve	s1m1	R	S	S	S	R	S
18	A147C(1)	Clinical Isolate	23.10.2019	56/M	Mild Gastritis	+ve	s1m1	R	S	S	S	S	S
19	A200C(2)	Clinical Isolate	18.03.2020	50/F	FD	+ve	s1m1	R	S	S	S	S	S
20	A201C(2)	Clinical Isolate	18.03.2020	35/M	DU + Peptic Ulcer	+ve	s1m1	R	S	S	S	S	S
21	A202A(1)	Clinical Isolate	18.03.2020	36/M	DU	+ve	s1m1	R	S	S	S	R	S

Note:

MET = Metronidazole, DU = Duodenal, Ulcer R = Resistant.

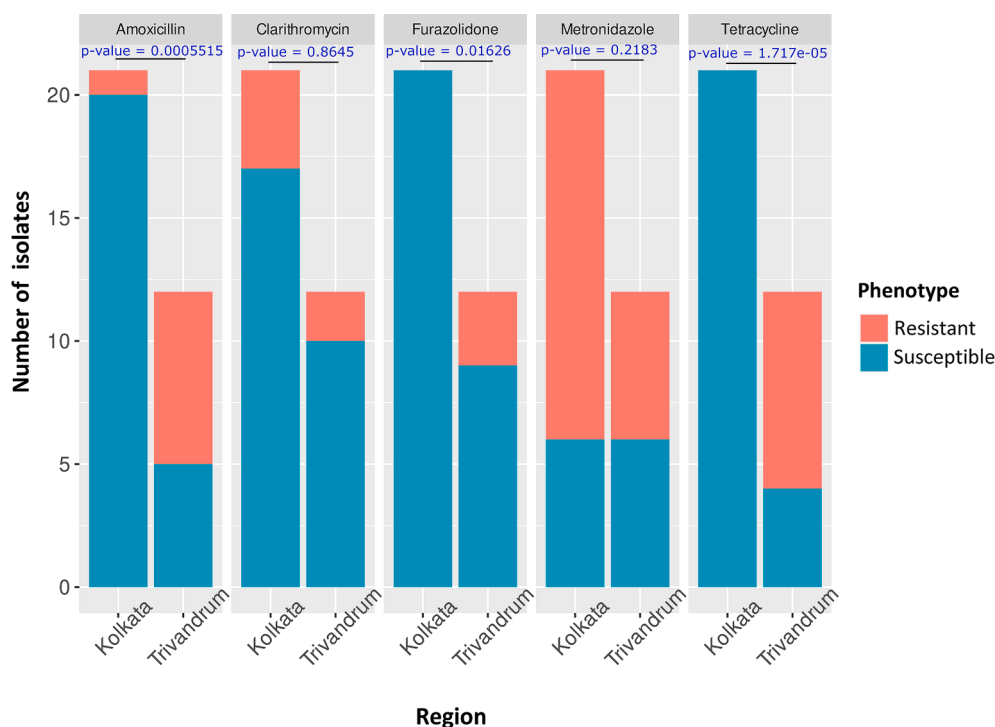
TET = Tetracycline, NUD = Non Ulcer Dyspepsia, S = Sensitive.

CLR = Clarithromycin, GERD = Gastroesophageal Reflux Disease, M = Male.

FUZ = Furazolidone, CA = Cancer, F = Female.

LEV = Levofloxacin, FD = Functional dyspepsia.

AMX = Amoxicillin.

**Fig. 1.** Comparison of AMR profiling of 20 isolates from eastern India (Kolkata) and 12 southern India (Trivandrum) by Pearson's Chi-squared test of categorical data. The y- and x- axis represent number and geographical locations of *H. pylori* isolates respectively.

twice. Such copy numbers of the gene may provide additional fitness to *H. pylori* (Supplementary Table 6). A total 6 types of major efflux pumps were observed in the *H. pylori* genome named as RND (resistance nodulation cell division), ABC (ATP binding cassette transporter), MATE

(multidrug and toxic compound extrusion), MFS (major facilitator superfamily), *hefB* (CmeE), *hefA* (CmeD) which were present in multiple copies shown in Supplementary Table 7. Since multidrug efflux systems majorly shows resistance against different antibiotics, we tried to

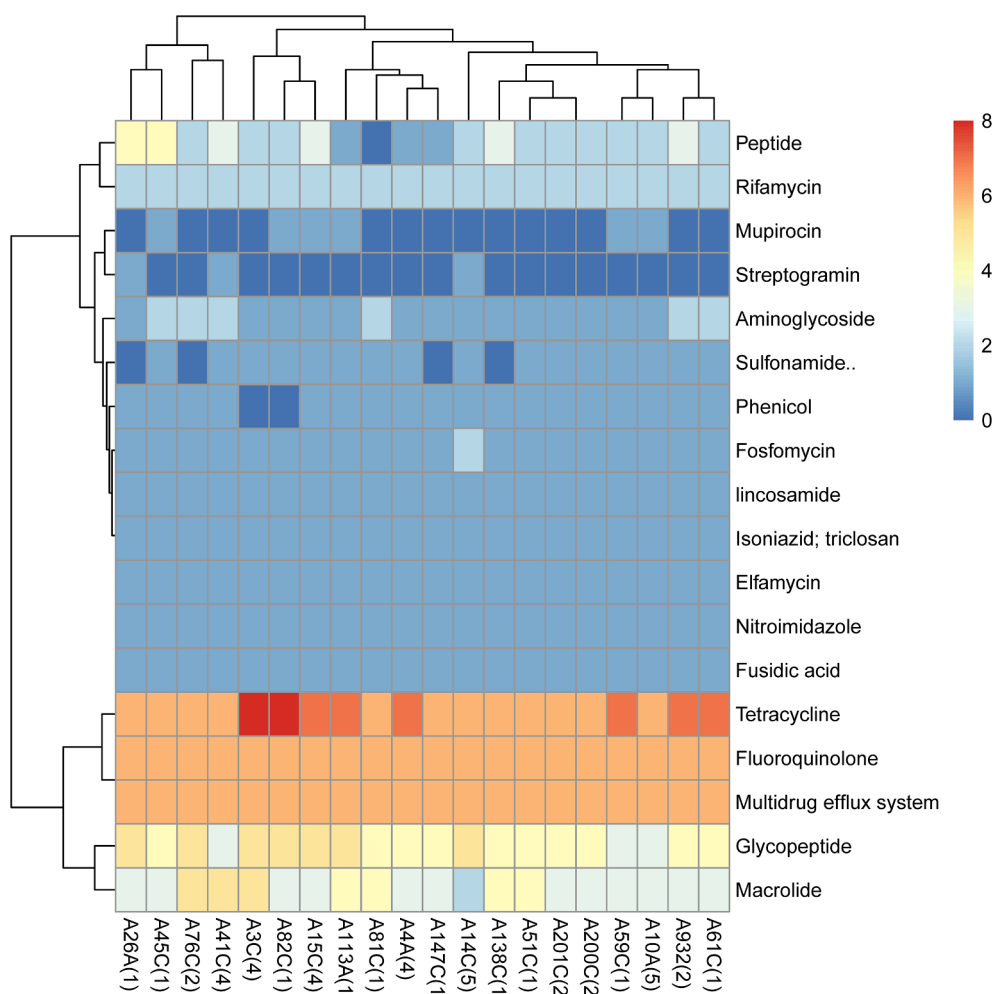


Fig. 2. Heatmap showing the copy number of antimicrobial resistant genes in each *H. pylori* genomes (n = 20) isolated from eastern India. The colour scale from blue (minimum) to red (maximum) represents the number of AMR genes in each *H. pylori* genomes for each drug class. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

compare the ORF and promoter sequences of all these 20 *H. pylori* isolates. When we compared it, we found no ORF sequence matching with the reference sequence ORF (*H. pylori* 26695) and only a few promoter sequences were matching. But only one predicted promoter sequence of RND pump was present in 9 isolates and a reference sequence and after aligning it we observed no mutations suggesting low antibiotic pressure on the organism. The comparison of promoter sequences of all 20 strains with the reference strain *H. pylori* 26695 against 50% sequence homology criteria has been depicted in **Supplementary Table 8**. An interesting finding was observed in the copy number of genes involved in virulence, disease and defense mechanism among the strains isolated at these two geographical locations as they show significant differences in mean value (mean of Trivandrum *H. pylori* isolates = 27.58333, mean of Kolkata *H. pylori* isolates = 15.05000, p-value (t-test) = 0.0005745). These results suggest the presence of high number of genes involved in virulence, disease and defense in *H. pylori* strains isolated from Trivandrum as compare to Kolkata this can be due to higher multidrug resistance observed in South Indian isolates.

2.4. Analysis of spontaneous mutations and acquired resistance genes in the genomes of *H. pylori*

To identify major cause of AMR resistance in *H. pylori* genome, we first analysed point mutations in the antibiotic's target gene, which contribute significantly in the emergence of resistant variants. Using

H. pylori 26695 as the reference strain, with known resistant and susceptible variants, we have aligned the sequences of antibiotic target genes *gyrA*, *gyrB*, *pbp1A*, *porD*, *OorD*, *16S rRNA* and *rdxA* (Mehrotra et al., 2021). We observed that four isolates A10A(5), A26A(1), A138C(1) and A82C(2) have decreased susceptibility to levofloxacin which carried a point mutation in the *gyrA* gene at position 87 (N87K). At 91st position, three different point mutations (D91G, D91N, D91Y) were detected in the same protein in the rest of the isolates (Table 2, Fig. 3). The results indicate that amino acid changes in the 87th and 91st positions are possibly responsible for the levofloxacin resistance in *H. pylori*. We have also analysed the protein sequences of *gyrB* from all the 20 isolates but no point mutations were found in them. The protein sequences of penicillin-binding proteins (*pbp1A*) are responsible for amoxicillin resistance in all the isolates. Analysis of *pbp1A* proteins reveals that 12 out of 20 strains carried a mutation in the *pbp1A* at 479th position (D479E). The same mutation was also present at 480th position (D480E) in A76C(2) and A14C(5) isolates. In addition, multiple amino acid substitutions were also detected in the strains at different positions (Table 2). In the case of *16S rRNA* gene sequences that confers tetracycline resistance, we observed 13 out of 20 strains carried nucleotide substitutions (Table 2). Metronidazole resistance in *rdxA* were studied in all 20 isolates, 12 out of 20 showed substitutions at 131st position (R131K). Seven isolates showed stop codons at several positions. In addition, at position 5 (D5N) substitution was observed only in A14C(5) strain (Table 2). Furazolidone resistance targeting *porD*, *OorD* were

Table 2

Mutations in various genes contributing resistance to Levofloxacin (LEV), Amoxicillin (AMX), Metronidazole (MET), Furazolidone (FUZ) and Tetracycline (TET) in *H. pylori* isolates.

Isolates	AMR genes						
	Levofloxacin (LEV)		Amoxicillin (AMX)	Metronidazole (MET)	Tetracycline (TET)	Furazolidone (FUZ)	
	<i>GyrA</i>	<i>GyrB</i>	<i>pbp1A</i>	<i>rdxA</i>	<i>16s rRNA</i>	<i>porD</i>	<i>oorD</i>
A76C(2)	Asp91/ Gly	–	Asp480/ Glu, Asn505/ Asp, Met516/ Ile	–	–	–	C156/T
A81C(1)	Asp96/ Asn	–	Asn504/ Asp, Met515/ Ile	Arg131/ Lys, Gln146/ stopcodon	A996/G	–	C156/T, C165/T
A59C(1)	Asp91/ Gly	–	Asp479/ Glu, Met515/ Ile, 595Gly, 596Gly, Lys650/ Gln, Arg651/ Lys	–	A933/ C, A996/ G,	–	C156/T, C165/T
A10A(5)	Asn87/ Lys	–	Asp479/ Glu, Asn504/ Asp, Met515/ Ile, Lys648/ Gln, Arg649/ Lys	Arg131/ Lys, Lys60/ stopcodon	T1003/ C,	–	C156/T, C165/T
A41C(4)	–	–	Met516/ Ile	Arg131/ Lys	–	–	C165/T
A14C(5)	Asp91/ Tyr	–	Lys649/ Gln, Arg650/ Lys	Asp5/ Asn, Gln119/ stopcodon	A996/ G, T1003/ C	–	A78/ G, C156/ T, C165/ T
A26A(1)	Asn92/ Lys	–	Asp479/ Glu, Asn504/ Asp, Met515/ Ile Lys648/ Gln, Arg649/ Lys	Arg131/ Lys	A996/ G, T1003/ C	–	C156/T, C165/T
A113A(1)	Asp91/ Tyr	–	Lys648/ Gln, Arg649/ Lys	Arg131/ Lys	A996/ G, T1003/ C	–	C156/T
A201C(2)	–	–	Asp479/ Glu, Met515/ Ile Lys648/ Gln, Arg649/ Lys	Arg131/ Lys	A996/ G, T1003/ C	–	A78/ G, C156/ T, C165/ T
A138C(1)	Asn87/ Lys	–	Asp479/ Glu Lys648/ Gln, Arg649/ Lys	Arg131/ Lys	–	–	C156/T, C165/T
A82C(2)	Asn87/ Lys	–	Asp479/ Glu, Met515/ Ile Lys648/ Gln, Arg649/ Lys	Arg131/ Lys	–	–	A78/ G, C156/ T, C165/ T
A93C(2)	Asp91/ Gly	–	–	Arg131/ Lys Gln146/ stopcodon	–	–	–
A45C(1)	Asp91/ Gly	–	Asp479/ Glu, Met515/ Ile Lys648/ Gln, Arg649/ Lys	–	–	–	A78/ G, C165/ T
A15C(4)	Asp91/ Asn	–	Asp479/ Glu, Met515/ Ile Lys648/ Gln, Arg649/ Lys	Gln50/ stopcodon	A996/ G	–	A78/ G, C156/ T, C165/ T
A200C(2)	–	–	Asp479/ Glu, Met515/ Ile Lys648/ Gln, Arg649/ Lys	Arg131/ Lys	A996/ G, T1003/ C	–	A78/ G, C156/ T, C165/ T
A51C (1)	Asp91/ Gly	–	Asp479/ Glu, Asn504/ Asp Met515/ Ile, Lys648/ Gln, Arg649/ Lys	–	A996/ G, T1003/ C	–	C156/ T, C165/ T
A61C(1)	Asp91/ Tyr	–	Asp479/ Glu, Asn504/ Asp Met515/ Ile	Gln50/ stopcodon	A996/ G, T1003/ C	–	A78/ G, C156/ T
A4A(4)	Asp96/ Asn	–	–	Arg131/ Lys	A998/ G, T1005C	–	–
A3C(4)	–	–	Asp479/ Glu, Met515/ Ile Lys648/ Gln, Arg649/ Lys	Gln130/ stopcodon	A996/ G, T1003/ C	–	A78/ G, C156/ T, C165/ T
A147C(1)	Asp91/ Gly	–	–	Arg131/ Lys	A996/ G	–	C156/ T, C165/ T

Note:

(-) = No mutation.

**Helicobacter pylori* 26695 is used as the reference strain for detection of point mutation.

analysed, in case of *porD* gene no point mutations were found in the *H. pylori* isolates. Various substitution mutations were detected in the *OorD* gene in all the 20 isolates taken in this study (Table 2, Fig. 3). To decode linkage between AMR genes and MGEs (IS elements, Phage, Integrase/recombinase, and Transposon/transposase). We searched for the presence of AMR genes in the close vicinity of MGEs and also checked GC content variations between AMR genes and MGEs if located within ≤ 10 Kb. However, we did not find any linkages between predicted AMR genes and the MGEs.

3. Discussion

In this study, we analyzed the AMR profile of *H. pylori* strains isolated from eastern India (Kolkata) and compared it with the southern isolates reported in our previous study (Mehrotra et al., 2021). Our comparative study depicted that the resistance profiles of *H. pylori* isolated from Kolkata against tetracycline, furazolidone, and amoxicillin are lower compared to the *H. pylori* strains isolated from South India at phenotypic level. At, molecular level we observed more AMR genes along with spontaneous mutations at 91st position in the QRDR region of *gyrA*, whereas no mutation observed for *gyrB*, low mutation for *pbp1A* gene as compared to southern isolates (Table 2). Metronidazole and tetracycline

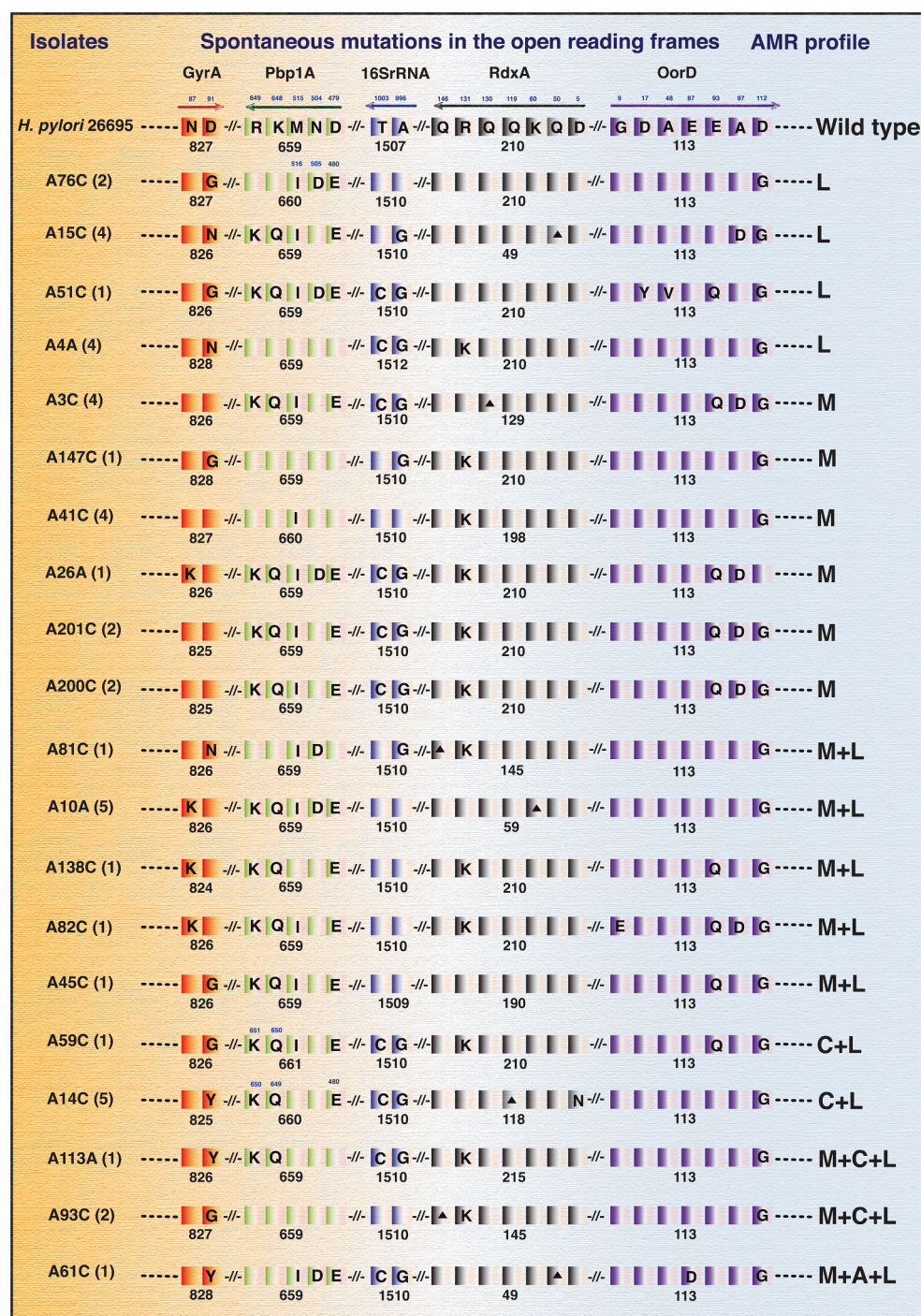


Fig. 3. Diagram (not to scale) depicting spontaneous mutations in different antimicrobial resistant genes in *H. pylori* strains isolated from eastern India. All the ORF of isolates associated with AMR is compared with the reference strain (*H. pylori* 26695). Each arrow indicates transcription direction. Size of ORF is shown via amino acid number or base pair. Codon number is mentioned on the top of all the mutations. Also, resistance encoded by each isolate is compared with the mutations obtained after aligning the sequences. Symbol '▲' represents deletion.

antibiotics have the same pattern of mutations between the two regions. We further found that southern isolates were more resistant to furazolidone as compared to eastern ones since no point mutations were detected in the *porD* gene of *H. pylori* isolates from eastern India (Table 2). Since *H. pylori* strains isolated from two different regions of India have not yet shown signs of acquired resistance. This signals less antibiotic pressure on the organism and less opportunities for horizontal gene transfer (HGT), but it is still feasible that the organism could mutate and evolve in the future while earning a fitness benefit.

One of the well-studied pathogenicity islands of *H. pylori* is the *cagPAI*. Approximately, 27 to 32 genes have been reported in the *cagPAI*, in which *cag* specific type IV secretion system (T4SS) and a secreted effector protein (*CagA*) are encoded, that are considered major risk

factors for the development of peptic ulcer disease and gastric adenocarcinoma (Cover et al., 2020). The dN/dS ratio of 18 *cag* genes of *cagPAI* are significantly (p-value < 0.05) different in South and East Indian strains. There are five *cag* genes such as *cagC*, *caggamma*, *cagH*, *cagG* and *HP0521* and that have shown distinct diversity (p-value < 10^{-10}) among these 18 statistically significant *cag* genes. The most interesting observation was found in the case of *vacA* gene though dN/dS ratio is < 1. Its dN/dS ratio has shown a higher difference (p-value < 10^{-16}) between East and South Indian isolates (Supplementary Information, Supplementary Fig. 2). It is well-known that the *VacA* toxin (encoded by *vacA* gene) enhances the ability of *H. pylori* to colonize the stomach and contributes to the pathogenesis of gastric adenocarcinoma and peptic ulcer disease (Foegeding et al., 2016). We observed in this

study that the genes involved in virulence, disease, and defense mechanisms have significant differences in South and East Indian isolates. Many studies also indicate that geographical and socio-cultural differences have an impact on *cagPAI* genetic diversity that leads to disease development and severity (Azuma et al., 2004, Akhter et al. 2007, Olbermann et al., 2010, Yuan et al., 2017). In pan-genome analysis an increase of 554 gene clusters was identified from a previous study (Mehrotra et al., 2021). This change was because of the addition of 20 *H. pylori* isolates from Kolkata. Also, core and softcore genomes have shrunk whereas shell and cloud subsets have expanded (Supplementary Information). This can be due to additional genes acquired by *H. pylori* strains isolated from eastern India.

4. Conclusion

In this study, 74 whole-genome sequences of *H. pylori* strains isolated from southern and eastern India were analysed. Twenty of the 74 genomes were from Kolkata, and their sequences were compared to the genomes of 12 previously reported *H. pylori* isolates from southern India. Our research revealed that the *H. pylori* strains isolated from eastern and southern India have significantly different patterns of AMR. In the East Indian (Kolkata) *H. pylori* isolates, we found lesser phenotypic and molecular resistance to amoxicillin and furazolidone antibiotics but higher phenotypic resistance to metronidazole and clarithromycin. Additionally, more AMR genes were found at the molecular level than in *H. pylori* isolates from southern India. From our findings, we suggest that metronidazole and clarithromycin antibiotics should be used judiciously in eastern Indian region. Based on our findings, spontaneous mutations are the major driver for AMR in *H. pylori* as till now, no acquired resistance has been observed. These mutations seem crucial for the emergence of MDR *H. pylori* strains, which needs further investigations. Significant changes in single nucleotide variations in more than 50% of *cag* genes of *cagPAI* may have a role in virulence diversity and severity in *H. pylori* infection which also reflect the adaptability of *H. pylori* in various geographical locations and socioeconomic conditions. The pan-genome research revealed that while the number of genes in the core and softcore subsets is decreasing, the genomes of the shell and cloud are expanding. It suggests that *H. pylori* genomes are plastic in nature. Additionally, we discovered a statistical difference between *H. pylori* strains isolated from the southern and eastern regions of India in the shell subgroup. With improved resolution, our core genome SNP-based phylogeny has been able to distinguish distinct strains of *H. pylori* based on geographical locations. The findings of the present study shed insight on the genomic variations, resistance phenotypes and genotypes, pathogenicity, and geographical association of *H. pylori* heterogeneity. The results of the current investigation can also be used to design nucleic acid probes to find potential resistance functions in *H. pylori* and the selection of appropriate antibiotic regimens in clinical settings, particularly in eastern and southern India.

5. Materials and methods

5.1. Patient's screening and processing of clinical samples

The patients included in this study (Table 1) visited the AMRI Hospital, Salt Lake, Kolkata with abdominal discomfort and were diagnosed with various gastric diseases ranging from gastritis, peptic ulcer to gastric cancer. Gastric biopsy specimens were collected in Brucella broth containing 25% glycerol and transported in cold condition to the National Institute of Cholera and Enteric Diseases, Kolkata for the isolation of *H. pylori*. Clinical *H. pylori* strains were isolated as previously described (Mehrotra et al., 2021). A single colony of *H. pylori* was picked up and propagated as a pure culture. The strains were identified as *H. pylori*, based on colony morphology, Gram-staining and biochemical tests like urease, catalase and oxidase and preserved in 20% glycerol at -80°C .

5.2. Evaluating antimicrobial susceptibility profile of different *Helicobacter pylori* isolates

To evaluate antimicrobial susceptibilities of the *H. pylori* strains, the Minimum Inhibitory Concentrations (MIC) were determined using agar dilution method for 5 antibiotics (metronidazole, clarithromycin, furazolidone, amoxicillin and tetracycline) that are commonly used in anti-*H. pylori* therapy. One reference American Type Culture Collection strain (ATCC 43504) was used for this study as a control. According to guidelines as described by the European Committee on Antimicrobial Susceptibility Testing (EUCAST) guidelines (EUCAST v.10; January 2020). *H. pylori* isolates were considered to be resistant when the MIC was $> 8\text{ }\mu\text{g/mL}$ for metronidazole, $>1\text{ }\mu\text{g/mL}$ for furazolidone, $>1\text{ }\mu\text{g/mL}$ for tetracycline $> 0.5\text{ }\mu\text{g/mL}$ for clarithromycin, $>1\text{ }\mu\text{g/mL}$ for levofloxacin and $> 0.125\text{ }\mu\text{g/mL}$ for amoxicillin. BHI Agar (BHIA) plates containing different concentrations of each antibiotic were prepared by adding the appropriate amount of the antibiotics from their respective stock solutions. All the antibiotics were purchased from Sigma Chemicals, USA. Frozen *H. pylori* cultures were revived on BHIA media and incubated under microaerobic conditions at 37°C for 3 days. Then culture was streaked on a BHIA plate and incubated for 1 day. The resulting young exponentially growing cells were suspended in Phosphate Buffered Saline (PBS), serial 10- fold dilutions of these suspensions were prepared and 10 μL of each dilution was spotted on freshly prepared antibiotic supplemented as well as on antibiotic free plate used as a control. All plates were incubated for 72 hrs. A strain was considered to be susceptible to concentrations of drugs that decreased its efficiency of colony formation by at least 1000-folds as compared with the control plate.

5.3. Whole genome sequencing

Nanodrop 2000 (Thermo Fisher, USA) was used to determine the concentration of the genomic DNA and the integrity was checked by agarose gel electrophoresis. The genomic DNA was then screened for whole genome sequencing based on two criteria: first, its $\text{OD}_{260/280}$ should range between 1.8 and 2.0 and second, there should not be any visual RNA contamination present in the DNA. Sequencing was done using high-throughput Illumina MiSeq sequencing platform (Illumina, Inc., USA). For library preparation, we used approximately 100 ng DNA followed by the pair-end sequencing using the Nextera XT DNA Library preparation kit (Illumina, Inc., USA).

5.4. Genome assembly, quality assessment and annotations

To find genomic features of 20 *H. pylori* strains, *de novo* based whole genome assembling of raw reads was done. Quality of raw reads were checked by Trimmomatic (version 0.39) program (Bolger et al., 2014) and low quality (QC value < 20) and short reads (length $< 36\text{ bp}$) were removed by following command “java -jar Trimmomatic-0.39.jar PE -threads 3 fastq1.gz fastq2.gz trimmed1.fq untrimmed1.fq trimmed2.fq untrimmed2.fq SLIDINGWINDOW:4:20 MINLEN:36”. Next cleaned and high-quality raw reads were used for genome assembly using Unicycler assembly pipeline (Wick et al., 2017). The quality of assembled genomes was evaluated based on the sequence contamination and completeness by CheckM (Parks et al., 2015) and QUAST (Gurevich et al., 2013) tools. Finally, all assembled genomes were annotated by the Rapid Annotation using Subsystem Technology (RAST) version 2.0 server (Aziz et al., 2008). The genome of *H. pylori* 26695 strain (Genome assembly ID: GCF_000008525.1) was used as a reference genome in this study (Tomb et al., 1997).

5.5. Antimicrobial resistance analysis of *Helicobacter pylori* isolates

We followed CARD (Comprehensive Antibiotic Resistance Database) (Alcock et al., 2020) and ISfinder to locate AMR genes and mobile

genetic elements (MGEs) (Siguier et al., 2006) respectively. Besides this, some MGEs were manually searched by using keywords such as IS elements, Phage, Integrase/recombinase, and Transposon/transposase etc. Linkage between AMR genes and MGEs was analysed as described in our previous finding (Mehrotra et al., 2021). For checking the possible mutations in the efflux pumps ORF and promoter sequences. We did prediction of promoters against these ORF by using an online available tool Neural network promoter prediction 1999 version 2.2 (Berkeley Drosophila Genome Project) (https://www.fruitfly.org/seq_tools/promoter.html). Promoters with a minimum score of ≥ 0.9 was selected for further analysis. In continuation, ORF and predicted promoter sequences were compared with the reference sequence (*H. pylori* 26695). Also, *H. pylori* resistant and susceptible genome sequences were downloaded from PATRIC 3.6.9 database (Wattam et al., 2014) and NCBI. Alignment was done by using the Clustal Omega tool (Sievers and Higgins 2014; Madeira et al., 2019) to find spontaneous mutations in the genome.

5.6. Nucleotide sequence submission

Whole genome sequences of all the 20 *H. pylori* strains are submitted to the NCBI GenBank database (SUB11093559). Sequences will be released after acceptance of the article.

5.7. Ethical clearance

The Human Ethics Committees of AMRI Hospital, Saltlake, Kolkata have approved this study. Written informed consent was obtained from each patient.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author contributions

Conceptualization, Funding acquisition, B.D.; Investigation, Methodology, Validation, SK, TM, DT, JV, BCK, SP, SC, AKP, SB, SK, MC, AP, SPK, BD; Project administration, Resources, AM, SB, AR, TRM, AKM, BD; Writing original draft, SK, TM, DT, AKM, and BD; Supervision, Software, Visualization, SC, AKM, BD; Writing - review and editing, AKM and BD.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gene.2022.146857>.

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